Miguel L. Rodrigues

Computer Vision & Control Systems Engineer

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Mechatronics engineer working at the intersection of **industrial computer vision** and **control theory**. I design, train, and deploy deep-learning inspection systems in production, and research event-triggered control of LPV systems. Published at ROCOND, SBAI, and CBA.

EXPERIENCE

AI / Computer Vision Engineer — Oct 2024 – Present

Digimet Solutions — Joinville, SC (remote)

- Build production deep-learning systems for automated **weld-bead inspection** — segmentation and keypoint-detection models feeding geometric measurement.
- Own the full pipeline: annotation infrastructure (Label Studio, EC2, PostgreSQL, S3), training, validation, mask post-processing, and deployment.
- Deploy **serverless inference on AWS Lambda** with CloudWatch monitoring; cut latency and cost via ONNX export and INT8 quantization, with domain adaptation to bridge training and field data.

Undergraduate Researcher (scholarship & volunteer) — Feb 2021 – Mar 2024

CEFET-MG — Departments of Mechatronics & Computer Engineering

- Controller design and Lyapunov stability analysis for linear systems; AI-based control of robotic manipulators (KUKA youBot).
- Fuzzy systems for financial-investment recommendation, ML for virtual drug screening, and image-capture systems for prostate-brachytherapy automation.

Teaching Assistant — 2024

CEFET-MG — Signals & Systems; Control Theory

RESEARCH & PUBLICATIONS

Event-triggered control of uncertain and nonlinear plants — LPV models, robust control under actuator saturation, LMI-based design, and Lyapunov stability.

- Co-Design and Evolutionary Optimization of Periodic Event-Triggered Robust Control for LPV Systems. Oliveira, **Rodrigues**, Guelton, Motchon, Leite. — ROCOND 2025
- Gerador Dinâmico de Eventos com Tratamento de Ruído para Redução de Transmissões. **Rodrigues**, Silva, Leite. — SBAI 2025
- Gerador de Eventos para Minimização de Transmissões em Sistemas LPV Saturantes. **Rodrigues**, Silva, Leite. — CBA 2024
- Modelagens e Simulações Computacionais Aplicadas ao Desenvolvimento Remoto de Sistemas Mecatrônicos. Silva, Alves, **Rodrigues**, Dâmaso. — CBA 2022

SELECTED PROJECTS

robotic_tools
Symbolic/numerical analysis of serial manipulators from Denavit–Hartenberg parameters: forward/inverse kinematics, Euler–Lagrange dynamics, gradient-based and evolutionary IK solvers.

prometheus
ESP32 closed-loop temperature controller with MQTT telemetry (with UFMG's GREA group): discrete PID with anti-windup, open-to-closed-loop transition, dual-core FreeRTOS.

beam_tools
Symbolic Euler–Bernoulli beam analysis with Macaulay singularity functions (SymPy): closed-form shear, moment, slope, and deflection for determinate and hyperstatic beams.

fpwm-signal-generator
ESP32 function generator (ESP-IDF/C): sine, sawtooth, triangle, and square waveforms via PWM and hardware timers, with under 1% frequency error.

EDUCATION

M.Sc., Electrical Engineering (in progress) — 2025 – 2027

CEFET-MG — Event-triggered control of LPV systems with actuator saturation

B.Sc., Mechatronics Engineering — 2020 – 2025

CEFET-MG — Control systems, automation, robotics, embedded systems

SKILLS & LANGUAGES

VISION & AI	PyTorch · OpenCV · semantic & instance segmentation · keypoint detection · ONNX · quantization · domain adaptation
CLOUD & INFRA	AWS (Lambda, EC2, S3) · Docker · CloudWatch · CI
PROGRAMMING	Python · C · TypeScript
CONTROL & COMPUTING	LPV systems · event-triggered & robust control · LMIs · Lyapunov · NumPy · SciPy · SymPy
EMBEDDED	ESP32 · STM32 · ESP-IDF · FreeRTOS · KiCad · power electronics
LANGUAGES	Portuguese (native) · English (full professional) · French (elementary)